

LEGAL METROLOGY DETAILED TEST REPORT

NON-AUTOMATIC WEIGHING INSTRUMENT (NAWI) - TECHNICAL AUDIT RECORD

Regulatory Standard: OIML R 76-1:2006 | Evaluation Mode: TYPE_EVALUATION

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REPORT NO: R76-2026-000074

QR VERIFICATION ID: R76-2026-B76A044A

1. LABORATORY INFORMATION

Facility: Central Legal Metrology Testing Laboratory

Address: Central Metrology Directorate

Assigned Technician: Pramod Patil

2. ENVIRONMENTAL CONDITIONS

Ambient Temperature: 22.5 °C

Relative Humidity: 50 %

Atmospheric Pressure: 1013.25 hPa

Compliance: Within standard metrological tolerance limits

3. INSTRUMENT TECHNICAL SPECIFICATION

Manufacturer: Avery Weigh-Tronix

Model: ZM301

Serial No: AW-90214

Type: Non-Automatic Weighing

Accuracy Class: Class III

Max: 30 kg

Min: 0.2 kg

Instrument (Standard Bench Scale)

Verification (e): 0.01 kg

Resolution (n): 3,000 intervals

Unit: kg

Interval (d): 0.005 kg

Indication: Electronic Digital

4. DETAILED METROLOGICAL TEST AUDITS & CALCULATION TRACES

TEST: WEIGHING PERFORMANCE & INTRINSIC ERROR (WEIGHING PERFORMANCE) Reference: OIML R 76-1:2006 Clause 3.5.1 & A.4.1 STATUS: PASS

Formulae: $P = I + 0.5e - 9\Delta$ & $w \text{ Error } E = P - L$ | Corrected Error $E_c = E - E_0$

No.	Phase	Load L (kg)	Ind. I (kg)	dL	E0	Raw E	Corr. Ec	MPE (+/-)	Decision
1	LOADING	0.200	0.204	-	0.000	0.004	0.004	+/-	PASS
2	LOADING	1.000	1.004	-	0.000	0.004	0.004	+/-	PASS
4	LOADING	10.000	10.009	-	0.000	0.009	0.009	+/-	PASS
5	LOADING	25.000	25.012	-	0.000	0.012	0.012	+/-	PASS
5	LOADING	5.000	5.001	-	0.000	0.001	0.001	+/-	PASS

TEST: REPEATABILITY TEST (REPEATABILITY)

Reference: OIML R 76-1:2006 Clause 3.6.1 & A.4.2 STATUS: PASS

Evaluation Formula: $\Delta I = I_{\max} - I_{\min}$ | Applicable Limit: MPE at Test Load

RECORDED TEST READINGS (RUNS):

R1: 15.001 kg | R2: 15.002 kg | R3: 15.009 kg | R4: 15.007 kg | R5: 15.007 kg | R6: 15.000 kg | R7: 15.006 kg | R8: 15.000 kg | R9: 15.000 kg | R10: 15.004 kg

Test Load: 15.000 kg

Max Indication (I_{max}): 15.009 kg

Min Indication (I_{min}): 15.000 kg

Difference (Delta I): 0.009 kg

Applicable MPE: 0.010 kg

Decision: PASS

TEST: ECCENTRIC LOADING TEST (ECCENTRIC_LOADING) Reference: OIML R 76-1:2006 Clause 3.6.2 & A.4.3 STATUS: PASS

Evaluation: Test Load = 1/3 Max (10.000 kg) | Error at each position <= MPE

Position	Applied Load (kg)	Indication (kg)	dL	Raw Error E	Corrected Ec	MPE	Result
CENTER	10.000	10.005	-	0.005	0.005	±0.010	PASS
FRONT_LEFT	10.000	10.001	-	0.001	0.001	±0.010	PASS
FRONT_RIGHT	10.000	9.998	-	-0.002	-0.002	±0.010	PASS
REAR_LEFT	10.000	10.002	-	0.002	0.002	±0.010	PASS
REAR_RIGHT	10.000	10.007	-	0.007	0.007	±0.010	PASS

TEST: ZERO-SETTING & ZERO-TRACKING EVALUATION (ZERO SETTING)

Reference: OIML R 76-1:2006 Clause 4.5 & A.4.2 STATUS: PASS

Initial Indication: 0.000 kg

Changeover 9D 0.004 kg

Calculated Zero Error E0: 0.001 kg

Applicable Limit: ±0.25 e (±0.003 kg)

Result: PASS

TEST: DISCRIMINATION TEST (DISCRIMINATION)

Reference: OIML R 76-1:2006 Clause 3.8 & A.4.8 STATUS: PASS

Requirement: Extra load of 1.4 d smoothly placed must step indication to I + d (Clause 3.8 / A.4.8)

Base Load (L): 15.000 kg

Initial Indication: 15.000 kg

Additional Load: 0.0007 kg (1.4 d)

Resulting Indication: 15.005 kg

Observed Shift (9D) 0.0050 kg

Mandated Shift: 0.0050 kg (1.0 d)

Decision: PASS

Note: Addition of 1.4d extra load (0.0007 kg) produced the mandated indication st

TEST: MULTIPLE INDICATING DEVICES COMPARISON (MULTIPLE INDICATING DEVICES)

Reference: OIML R 76-1:2006 Clause 3.6.3 STATUS: NOT_APPLICABLE

This test module is not applicable to the current instrument configuration.

TEST: DIFFERENT POSITIONS OF EQUILIBRIUM (DIFFERENT POSITIONS OF EQUILIBRIUM)

Reference: OIML R 76-1:2006 Clause 3.6.4 STATUS: NOT_APPLICABLE

This test module is not applicable to the current instrument configuration.

OVERALL VERIFICATION RESULT: PASSED (COMPLIANT)

All 5 applicable OIML R 76-1:2006 metrological tests have been verified compliant with Maximum Permissible Errors.

PUBLIC VERIFICATION RECORD & TRACEABILITY

Permanent Verification ID: R76-2026-B76A044A

Authentic URL to inspect full measurement calculations:

<http://localhost:5173/public/report/R76-2026-B76A044A>

OFFICIAL ATTESTATION & SIGN-OFF

Officer: Pramod Patil

Facility: Central Legal Metrology Testing Laboratory

[DIGITALLY RECORDED & ARCHIVED]

